OSMAN GONI

Software Engineer (AI/ML)

Deep Learning · Computer Vision · LLM Applications

Dhaka, Bangladesh
Phone: 01872-318280
Email: osmancgoni@gmail.com
LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI/ML-focused Software Engineer with a B.Sc. in Computer Science and Engineering from American International University-Bangladesh, specializing in deep learning, computer vision and applied LLM systems. Built production-style systems with Python, FastAPI, C#, ASP.NET, SQL Server and JavaScript, including a RAG-based PDF Q&A application using vector search. Contributed to computer vision research on U-Net architectures for medical image segmentation, currently under review for publication. Strong in data structures and algorithms, with 500+ Codeforces problems solved (rating 1109). Seeking an AI/ML Engineer role to build and deploy reliable machine learning systems, backed by solid full-stack engineering skills.

TECHNICAL SKILLS

- **AI / ML:** PyTorch, Deep Learning, Computer Vision, Image Segmentation (U-Net), Retrieval-Augmented Generation (RAG), LlamaIndex, Gemini API, Embeddings, Semantic Search
- **Languages:** Python, C++, C#, SQL, JavaScript
- **Backend & Frameworks:** FastAPI, ASP.NET, RESTful APIs, Node.js
- **Databases:** Microsoft SQL Server, Qdrant (vector database)
- **Tools & Workflow:** Inngest (workflow orchestration), Git, Debugging, Performance Optimization
- **Core Concepts:** Data Structures, Algorithms, SOLID Principles, Object-Oriented Programming

RESEARCH EXPERIENCE

Cervical Spine MRI Segmentation: U-Net Comparison | *PyTorch, Deep Learning, Computer Vision* 2025 – 2026

Research paper comparing Convolutional, Transformer and Hybrid U-Net models; currently under review for publication.

- Worked with the Duke spine MRI dataset, a fresh, high-resolution and unique dataset.
- Preprocessed the data to meet the input requirements of each model family.
- Trained and evaluated multiple segmentation architectures and compared performance metrics to identify the best-performing models.

PROJECTS

RAG PDF Analyzer | *Python, FastAPI, LlamaIndex, Gemini, Qdrant, Inngest*

- Built a Retrieval-Augmented Generation (RAG) PDF question-answering system with a FastAPI backend and Gemini as the LLM.
- Implemented PDF chunking and embedding generation using LlamaIndex and Gemini embeddings.
- Integrated Qdrant as the vector database to enable fast semantic search over document content.
- Used Inngest for workflow orchestration to make document processing reliable and fault tolerant.

Hospital Management System | *C#, ASP.NET, RESTful APIs, Microsoft SQL Server*

- Designed and implemented RESTful APIs in C# and ASP.NET with a clean, well-organized code structure.
- Used Microsoft SQL Server for database management and persistent data storage.
- Applied SOLID principles to improve maintainability and scalability of the codebase.

EDUCATION

Bachelor of Science in Computer Science and Engineering | American International University-Bangladesh

COMPETITIVE PROGRAMMING

- Solved 500+ problems on Codeforces and 1000+ competitive programming problems overall.
- Codeforces rating: 1109. CodeChef: 2-star (1405).
- Regularly compete in programming contests, building algorithmic thinking and the ability to solve complex problems under time pressure.